Duc Hoang

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Summary

High-energy physicist specializing in machine learning, data science, and real-time FPGA systems. Extensive experience delivering production-level ultra-low-latency ML inference on FPGAs and petabyte-scale data analysis for CERN's CMS experiment. Proven leader in multi-institutional projects, skilled in developing custom ML solutions and deploying robust, high-performance software.

Education

- **Massachusetts Institute of Technology (MIT)** — *2021 - 2026*
  PhD, High Energy Physics — Cambridge, MA
- **Rhodes College** — *2017-2021*
  B.S., Physics, Summa Cum Laude — Memphis, USA

Professional Experience

- **MIT & European Organization for Nuclear Research (CERN)** — *2021 - Now*
  Graduate Researcher & Next Generation Trigger PhD Fellow — Cambridge, USA & Geneva, Switzerland
  - Coordinated multi-institutional team to develop **ultra-low-latency FPGA-accelerated ML models** for CMS Level-1 Trigger, which processes approximately 10% of global internet traffic in real time. Hear me explain it along the trigger leardership team **[1]**
  - Built first **FPGA-based ML algorithms for real-time jet tagging** and delivered cross-platform, low-latency data transfer tests between different FPGA board platforms. **[2]**
  - Analyzed all **Run 2 CMS data (a few PBs)** to deliver **most sensitive Higgs $\rightarrow b\bar{b}$ measurement in the V(qq)H(bb) decay channel**, invited presentation at Rencontres de Moriond 2025. **[3]**
  - Created and taught data analysis projects for **MIT 8.16/8.316: Data Science in Physics**, which is published on **MITx** course **Computational Data Science in Physics**.
- **Fermi National Accelerator Laboratory** — *2019 - 2021*
  Undergraduate Researcher — Chicago, USA
  - Developer of hls4ml, A package for **ultra-low-latency machine learning inference in FPGAs** using high level synthesis language (HLS). **[4]**
  - Designed superconducting magnet quench detection systems **[5]**
  - Analyzed simulation data to predict potential signal gains for the real-time trigger systems for Light Dark Matter Experiment (LDMX). **[6]**
  - Machine Learning analysis to infer hyper-parameter optimization performance of models used in the MINERvA Experiment. **[7]**

Selected Publications & Software

[1] **CMS Collaboration**, "The Phase-2 Upgrade of the CMS Level-1 Trigger", https://cds.cern.ch/record/2714892

[2] **CMS Collaboration**, 2025. "Multiclass object tagging in the CMS Phase-2 Level-1 Correlator Trigger", https://cds.cern.ch/record/2936315.

[3] **CMS Collaboration**, "Search for a boosted Higgs boson decaying to bottom quark pairs in association with a hadronically decaying W or Z boson with the CMS detector using proton-proton collisions at $\sqrt{s} = 13$ TeV", https://cds.cern.ch/record/2928061

[4] `hls4ml` **Community**, "hls4ml: An Open-Source Codesign Workflow to Empower Scientific Low-Power Machine Learning Devices"

[5] **D. Hoang** et al., "Intelliquench: an adaptive machine learning system for detection of superconducting magnet quenches.," in *IEEE Transactions on Applied Superconductivity*, doi: 10.1109/TASC.2021.3058229.

[6] **LDMX Collaboration**, "Current Status and Future Prospects for the Light Dark Matter eXperiment" [arXiv:2203.08192[hep-ex]].

[7] **D. Hoang** et al. "Inferring Convolutional Neural Networks' Accuracies from Their Architectural Characterizations," *2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA)*, Boca Raton, FL, USA, 2019, pp. 1388-1391. DOI: 10.1109/ICMLA.2019.00226

Skills

- **Specialized Area:** Big Data Analysis, Statistical Modeling, ultra-low latency FPGA Inference, Machine Learning
- **Programming Languages:** Python, C++, HTML, CSS, TypeScript
- **Web Technologies:** Django, Angular
- **Database Systems:** SQLite, PostgreSQL
- **Data Science & Machine Learning:** Most Python-related machine learning tools.
- **Cloud Technologies:** Google Cloud
- **DevOps & Version Control:** Continous Integration, Docker, Singularity/Apptainer, Git

Honors and Awards

- **The 2025 Breakthrough Prize in Fundamental Physics** *2025*
  Received as a member of the CMS Collaboration [🌐]
  - The prize is awarded to physicists from theoretical, mathematical, or experimental physics that have made transformative contributions to fundamental physics, and specifically for recent advances.
- **Next Generation Trigger PhD fellow.** *2024*
  CERN [🌐]
  - Awarded for exceptional PhD students to work along CERN scientists to develop the next generation trigger system at the Large Hadron Collider.
- **MIT Public Service Fellow (PKG) Fellowship** *2025*
  Priscilla King Gray Public Service Center - MIT [🌐]
  - Founded and organized, along with other International Olympiad Winners from Central Vietnam, a STEM summer school, providing under-privileged high school students in Central Vietnam the opportunity to study emerging fields such as artificial intelligence, data science, programming, renewable energy, and quantum technology at no cost.
- **2024-25 Christiaan Huygens Graduate Fellowship - MIT.** *2025*
  MIT School of Science
  - Internal Fellowship providing financial support for the full academic year in the School of Science for students with non-traditional backgrounds.
- **Henry W Kendall (1955) first year graduate fellowship.** *2021*
  MIT Laboratory for Nuclear Science
  - Financial support for first year graduate students.
- **Miscellaneous Undergraduate Awards.** *2017-2021*
  Rhodes College
  - Research Award in Physics, Council on Undergraduate Research (CUR) Physics & Astronomy Student Travel Award, W.O Shewmaker Award for Excellence in First-Year Search Program, Award for Excellence in Physics by a First-Year Student, Summa Cum Laude, Sigma Pi Sigma, Phi Beta Kappa, Physics Honors.